PUBLIC CASE STUDY

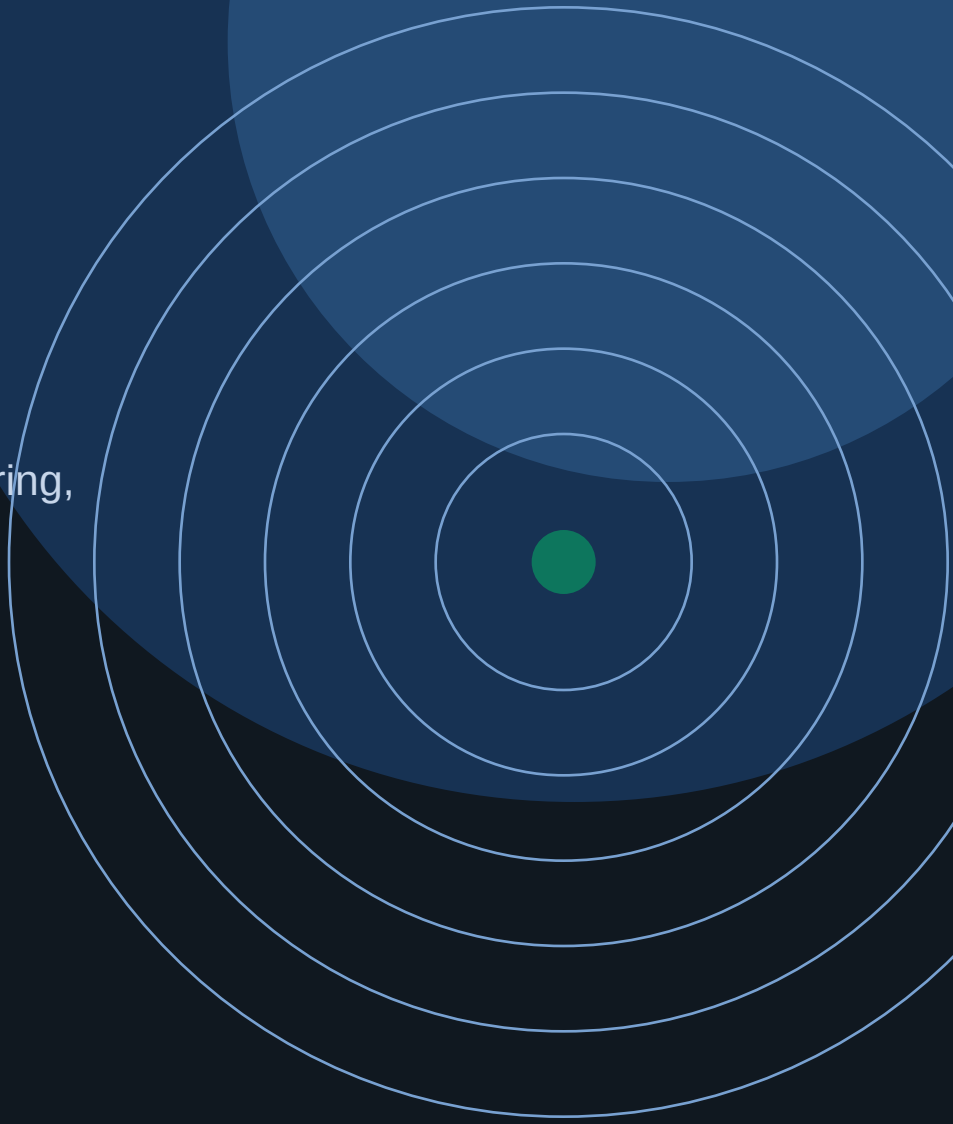
# Fraud Risk Decisioning

A technical case study in anomaly detection, risk scoring, and decision threshold optimization

ANOMALY DETECTION

PREDICTIVE MODELING

DECISION OPTIMIZATION



# A fraud system must manage risk without blocking legitimate activity.

This project examined a realistic decisioning problem: detect abnormal behavior, estimate application risk, and convert model output into actions that balance loss prevention with customer friction and review effort.

QUESTION 01

## Where is risk changing?

Use historical patterns to distinguish meaningful anomalies from ordinary variation.

QUESTION 02

## Which applications need attention?

Use predictive signals to rank risk while respecting the order in which data becomes available.

QUESTION 03

## How should the system respond?

Map risk scores to approve, review, or decline actions under operational constraints.

# One workflow connected diagnosis, prediction, and decision design.

The work treated fraud detection as an end-to-end analytics problem rather than a single modeling exercise. Each stage informed the next: patterns shaped features, evaluation reflected time, and predicted risk informed operational thresholds.

### 01 Diagnose

Clean case data, aggregate events over time, and isolate abnormal patterns.

### 02 Model

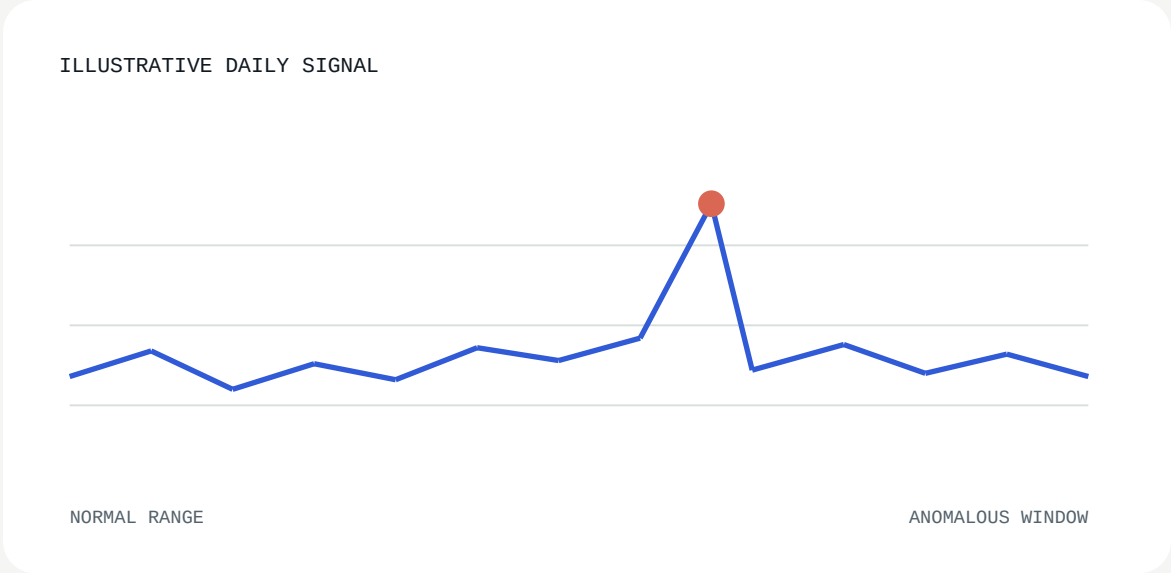
Build contextual features and compare risk models using chronological validation.

### 03 Decide

Optimize review and decline thresholds against a transparent business objective.

# Find deviations in context, then investigate what changed.

The first stage combined exploratory analysis with time-series reasoning. Daily application, approval, and fraud indicators established a baseline before the analysis drilled into product, geography, activity timing, and identity-related patterns.




### A Establish a baseline

Use seasonal time-series modeling to separate persistent deviations from expected cyclicality.

### B Trace contributing factors

Compare shifts across relevant business dimensions before forming a hypothesis.

# Rank risk with features that reflect behavior and context.

The predictive stage combined baseline features with derived signals. The design emphasized reproducible preparation, temporal ordering, and comparisons against interpretable and tree-based models.

FEATURE DESIGN

## Behavioral context

Represent time, activity patterns, location consistency, and group-level history without relying on a single raw field.

VALIDATION

## Chronological split

Train on earlier observations and validate on later observations to better approximate future decision conditions.

MODELING

## Compare before tuning

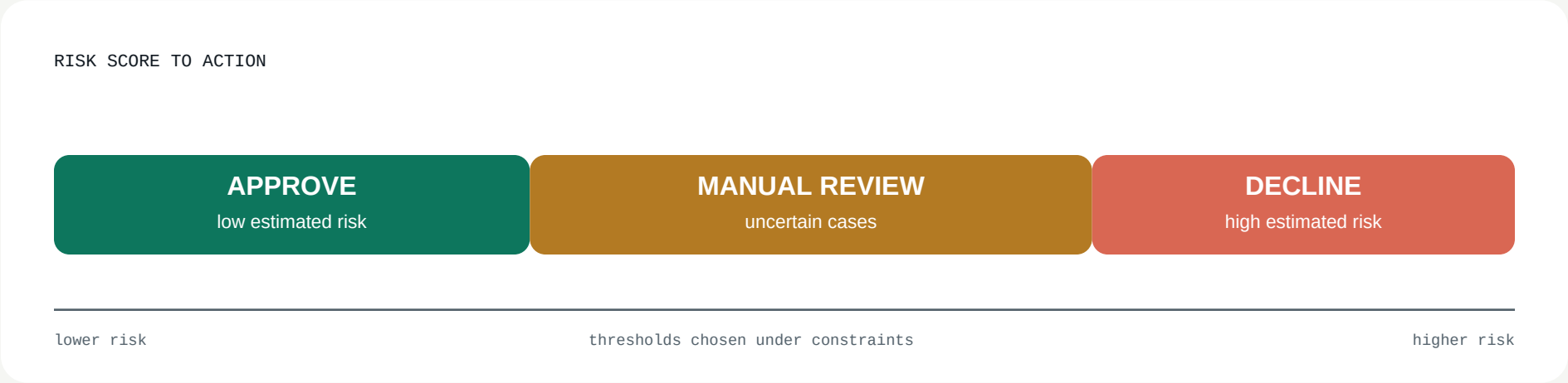
Start with a transparent baseline, then test non-linear tree-based models and tune only when the gain is justified.

METHODS USED

SARIMA · LOGISTIC REGRESSION · RANDOM FOREST · XGBOOST

# A useful risk model needs a decision policy, not just a score.

Model probabilities were translated into three operational paths. The thresholds were optimized against an explicit objective that accounted for expected value, fraud loss, manual review effort, and third-party data cost while preserving a workable review band.




Optimization treated policy design as part of the data science task: a model can rank risk well and still make poor decisions if its thresholds ignore real operational trade-offs.

# Technical modeling becomes more credible when it connects to a decision and its constraints.

INSIGHT

## Time matters

Anomaly detection and model evaluation should reflect the order in which information becomes available.

INSIGHT

## Context matters

Derived behavioral and historical signals can clarify the risk of a single application.

INSIGHT

## Operations matter

The final value of a model depends on how it changes a real decision workflow.

This public case study presents the project approach and technical reasoning. It intentionally omits source code, raw case data, vendor-specific details, exact performance figures, and financial assumptions.

SCOPE NOTE

Prepared as a portfolio case study. The material describes an analytical project and does not claim production deployment.